

AIOA Core Forensic Full Snapshot

Generated: 2026-05-24T18:25:09 Commit: 04adfbdb5a6b34d2969d67ac7e84c704c8e0915a

This PDF is a navigable forensic packet. Full raw source files are preserved separately in `source_export/` and subsystem markdown chunks.

AIOA Core Forensic Export

Generated: 2026-05-24T18:25:09

Repository: /home/l/Desktop/A0IA-Core

Checkpoint commit: 04adfbdb5a6b34d2969d67ac7e84c704c8e0915a

What This Export Is

This is a read-only forensic export snapshot of the AIOA Core application. It is intended for architecture auditing, external model review, forensic archival, reproducibility, future research analysis, and AI safety verification.

Repository State

```
## main...origin/main [ahead 1]
```

Latest commit:

```
04adfbdb (HEAD -> main) Checkpoint before forensic export snapshot
```

How To Navigate

Start here:

1. `architecture_summary.md` for system-level architecture.
2. `module_summaries.md` for module inventory.
3. `repository_tree.txt` for hierarchy.
4. `file_manifest.csv` for file-level metadata.
5. `chunked_markdown/` for subsystem-specific code review.
6. `source_export/` for full text/source file export preserving hierarchy.
7. PDFs for portable review.

Implemented vs Conceptual

Implemented:

- Runtime coordinator and bounded action loop.
- Structured execution engine with approval gate.
- Local RHCSA/Linux knowledge corpus.
- Canonical Linux PDF source and extraction.
- Candidate command index loader.
- Deterministic Linux retrieval engine v1.
- Retrieval tests.
- Memory/provenance doctrine and governance docs.

Conceptual or deferred:

- Promotion of candidate commands into canonical indexes.
- Feature-flagged runtime hook for the new retrieval engine.
- Unified retrieval facade across kernel, RHCSA search, and Linux retrieval engine.
- Production packaging for repeated forensic exports.

Known Risks

- Overlapping retrieval paths can diverge unless consolidated.
- Candidate corpus contains malformed/path/PDF artifact records and must not be blindly promoted.
- Runtime logs/state were part of the prior checkpoint; future policy should decide whether they remain versioned or move to archived artifacts.
- External providers are available but must remain lower priority than deterministic local evidence for Linux/RHCSA answers.

Export Contents

- repository_tree.txt
- file_manifest.csv
- architecture_summary.md
- module_summaries.md
- source_export/
- chunked_markdown/
- forensic_full_snapshot.pdf
- architecture_and_runtime.pdf
- retrieval_and_knowledge_layer.pdf
- memory_and_provenance.pdf

AIOA Core Forensic Architecture Summary

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Git status at export start:

```
## main...origin/main [ahead 1]
```

Latest commit:

04adfbdb (HEAD -> main) Checkpoint before forensic export snapshot

Runtime Flow

User input

- > local fast routes
- > external URL/repository boundary check
- > local deterministic knowledge route when applicable
- > model planning fallback
- > structured JSON action validation
- > human approval for non-response actions
- > local executor
- > operational memory/log update
- > final response or next bounded step

Retrieval Architecture

AIOA currently contains two related local retrieval/control paths:

- `runtime/adaptive_routing/epistemic_kernel.py`: deterministic epistemic kernel using RHCSA search, provenance, contradiction notices, pressure score, and routing depth.
- `runtime/retrieval/linux/`: first operational deterministic Linux retrieval engine with query normalization, exact/alias/subcommand/category/family/keyword lookup, scoring, provenance attachment, and refusal behavior.

The newer retrieval engine is tested but not yet wired into the main runtime router. That is intentional and avoids premature runtime behavior changes.

Provenance Model

Source lineage is represented through:

- `runtime/knowledge/manifests/library_manifest.yaml`
- `runtime/knowledge/provenance/PROVENANCE_POLICY.md`
- `runtime/provenance_registry.json`
- `runtime/contradiction_registry.json`

Canonical Linux source:

`runtime/knowledge/source/linux_master_library_v1.pdf`

SHA256: 7eab9450dd15cc5e1607c29d9fe3b19c4cf9854bb702f113534b6ec34a34dc03

Legacy source remains preserved:

`runtime/knowledge/source/RHCSA_Command_Library (1).pdf`

SHA256: b8092eeabbfd80489d9e5ce8b49ba4d822aa83cc360da0a8f3c76276ac21d6b7

Evidence and Reasoning Separation

The architecture documents define memory as layered authority, not one generic store:

- L0 ephemeral runtime state
- L1 operational logs
- L2 reasoning traces
- L3 provenance records
- L4 immutable evidence
- L5 contradiction registry

Important boundary: runtime logs and model reasoning must not become retrieval evidence without explicit source ingestion and provenance.

Deterministic Safeguards and Feature Flags

Runtime safeguards include:

- `EPISTEMIC_KILL_SWITCH`
- `EPISTEMIC_DISABLE_MODEL`
- `EPISTEMIC_DISABLE_KNOWLEDGE_ROUTE`
- `EPISTEMIC_DISABLE_MEMORY_HATS`
- `EPISTEMIC_DISABLE_REASONING_TRACE`
- `EPISTEMIC_DISABLE_UNKNOWN_FALLBACK`

The Linux retrieval engine itself refuses low-confidence queries below the deterministic confidence threshold and does not call external APIs, embeddings, vector databases, or autonomous loops.

Execution Boundaries

`runtime/tools/executor.py` dispatches structured actions only after validation. Non-response actions require human approval in normal runtime flow. Shell execution goes through command validation/classification before dispatch.

Candidate Promotion Pipeline

Current candidate parser statistics:

Metric	Count
total parsed entries	3152
total candidate records	3152
total unique candidate commands	2570
candidate-only entries	1978
duplicates against existing canonical/index	725

Metric	Count
internal candidate duplicates	582
malformed/unresolved entries	97

No candidate rows were promoted into canonical indexes during parsing. This is the correct safety posture.

Maturity Level

Current maturity: infrastructure prototype with strong local-first boundaries and an operational deterministic retrieval subsystem.

Implemented:

- bounded runtime loop
- approval-gated executor
- provider abstraction
- local RHCSA/Linux knowledge corpus
- canonical source manifest
- candidate parser and reports
- deterministic retrieval engine v1
- retrieval tests
- memory/provenance doctrine

Not yet implemented or intentionally deferred:

- runtime router hook for `LinuxRetrievalEngine`
- candidate promotion into canonical indexes
- reviewed alias/family expansion from candidate corpus
- full provider-independent retrieval answer renderer
- automated report packaging workflow inside repo

Known Limitations

- Retrieval paths overlap and should be unified behind one facade before router integration.
- Candidate data contains weak descriptions, path artifacts, and PDF merge artifacts.
- Runtime logs/state are present in the repository checkpoint and should receive a long-term archival/ignore policy.
- The Linux retrieval engine is intentionally not wired into the main route yet.
- The system has local-first retrieval but not a production-grade RAG/vector layer; this is by design for deterministic auditability.

Module Summaries

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Category Inventory

Category	Files	Bytes
configuration	10	21482
docs	50	164717
governance	26	301410
knowledge	88	6742375
memory	22	105073
provenance	6	47967
reports	1	5591
repository	20	45291
retrieval	6	19916
runtime	40	124382

Category	Files	Bytes
tests	10	47266
tooling	15	135684

Primary Modules

- `runtime/main.py`: runtime coordinator, local routes, model planning fallback, safeguards, session logging.
- `runtime/tools/executor.py`: structured action execution, approval gate, shell/filesystem/browser dispatch.
- `runtime/adaptive_routing/epistemic_kernel.py`: deterministic local epistemic control layer over RHCSA evidence.
- `runtime/retrieval/linux/`: deterministic Linux retrieval engine v1 with normalization, scoring, provenance attachment, refusal behavior.
- `runtime/knowledge/`: canonical RHCSA commands, command indexes, source PDF, extracted text, candidate index loader, reports.
- `runtime/memory/`: runtime state, evidence/reasoning trace helpers, RHCSA context injection.
- `docs/architecture/`: memory ontology, forbidden flows, access matrix.
- `MHLM_MHSR/`: governance/archive/taxonomy/case-study scaffolding for anti-hallucination analysis.

Repository Tree

AIOA-Core forensic repository tree
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```

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AOIA_CANONICAL_STRUCTURE_PLAN.md
AOIA_CONTAMINATION_REPORT.md
AOIA_DEPENDENCY_GRAPH.md
AOIA_MEMORY_ONTOLOGY.md
AOIA_RUNTIME_BOUNDARY_RECOMMENDATION.md
AOIA_RUNTIME_MAP.md
AOIA_TRANSITIONAL_COMPONENTS.md
AUTHORITY_SCOPE.md
CONTRADICTION_SEMANTICS.md
CURRENT_MEMORY_FLOW.md
FILESYSTEM_ONTOLOGY_LAYOUT.md
MEMORY_BOUNDARY_ANALYSIS.md
MEMORY_LAYER_DECOMPOSITION.md
MHLM_MHSR/
  case_studies/
    anti_hallucination_epi_app/
      README.md
    architecture/
      DEPENDENCY_BOUNDARY_ANALYSIS.md
      MEMORY_DOMAIN_SPLIT_PLAN.md
    contracts/
      RUNTIME_SAFETY_CONTRACTS.md
    enforcement/
      ENFORCEMENT_LAYER_DESIGN.md
  archive/
    AOIA_MASTER_LIBRARY/
      AOIA_Master_Library.pdf
      MASTER_INDEX.md
  contradictions/
    CONTRADICTION_TAXONOMY.md
  governance/

```

- GOVERNANCE_MODEL.md
- audit/
- authority/
- policies/
- review/
- risk_models/
- lineage/
- LINEAGE_POLICY.md
- decisions/
- events/
- sessions/
- prompts/
- PROMPT_ARCHIVE_POLICY.md
- normalized/
- raw/
- provenance/
- PROVENANCE_MODEL_PREP.md
- reports/
- architecture/
- forensic/
- governance/
- normalized/
- raw_provider/
- claude/
- MANIFEST.md
- codex/
- MANIFEST.md
- deepseek/
- MANIFEST.md
- gemini/
- MANIFEST.md
- kimim/
- MANIFEST.md
- unknown/
- MANIFEST.md
- synthesis/
- unclassified/
- UNCLASSIFIED_MANIFEST.md
- lsc_neutrino/
- README.md
- archive/
- contradictions/
- lineage/
- decisions/
- events/
- sessions/
- prompts/
- normalized/
- raw/
- provenance/
- reports/
- normalized/
- raw_provider/
- synthesis/
- docs/
- framework/
- governance/

- methodology/
 - contradiction_policy.md
 - evidence_policy.md
 - inclusion_rules.md
 - lineage_policy.md
- schemas/
 - artifact.schema.json
 - case_study_manifest.schema.json
 - lineage_event.schema.json
 - provenance_record.schema.json
 - report.schema.json
- taxonomy/
 - case_studies.yml
 - legacy_aliases.yml
 - model_aliases.yml
- imports/
 - git_bundles/
 - provider_exports/
 - normalized/
 - raw/
 - repo_snapshots/
- MUTABLE_STATE_ISOLATION_PLAN.md
- ORCHESTRATION_REMNANT_AUDIT.md
- PROVENANCE_FOUNDATION.md
- README.md
- ROADMAP.md
- ROUTING_AUTHORITY_ANALYSIS.md
- archive/
 - quarantine/
 - README.md
- contradictions/
 - README.md
- docs/
 - ADR/
 - ADR-001-deterministic-routing.md
 - ADR-002-three-depth-model.md
 - ADR-003-local-first-execution.md
 - ADR-004-no-runtime-learning.md
 - ADR-005-fail-fast-philosophy.md
- ARCHITECTURE.md
- CONSTRAINTS.md
- FULL_PROJECT_TREE.txt
- GIT_HISTORY_CONTINUATION_PLAN.md
- KNOWLEDGE_PACK_RULES.md
- KNOWLEDGE_PACK_SPEC.md
- LINEAGE_MAP.md
- LINUX_ENGINEERING_LIBRARY.md
- LINUX_ENGINEERING_LIBRARY_REPORT.md
- NON_GOALS.md
- PHASE1_COMPLETE_REPORT.md
- PHASE1_POSTCHECK.md
- PHASE1_STRUCTURE_REPORT.md
- PHASE2_DUPLICATION_SCAN.md
- PHASE2_MIGRATION_REPORT.md
- PHASE2_UNCLASSIFIED_ITEMS.md
- PHASE3_DEPENDENCY_RISKS.md
- PHASE3_GOVERNANCE_PREP_REPORT.md

- PHASE3_RUNTIME_PREP_STATUS.md
- PRE_PHASE1_CONFLICT_SCAN.md
- README.md
- REPOSITORY_CONSTITUTION.md
- REPOSITORY_STATE_REPORT.md
- REPO_STRUCTURE.md
- RHCSA_ENGINE_REVIEW.md
- RUNTIME_BOUNDARY.md
- TAXONOMY_NORMALIZATION_REPORT.md
- TEST_CONSTITUTION.md
- TRANSFER_CONTENT_REPORT.txt
- adr/
 - 0001-keep-aoia-isolated.md
 - 0002-minimal-deterministic-router-skeleton.md
 - 0003-immutable-startup-configuration.md
 - 0004-stdout-only-plain-text-logging.md
 - 0005-test-constitution-determinism-first.md
 - README.md
- architecture/
 - AOIA_MEMORY_MODEL.md
 - FORBIDDEN_MEMORY_FLOWS.md
 - MEMORY_LAYER_ACCESS_MATRIX.md
- checkpoints/
 - 2026-05-23/
 - AOIA_DAILY_CHECKPOINT.md
 - NEXT_ACTIONS.md
- forensic-runtime-audit/
 - CANONICAL_REFACTOR_PREP.md
 - CURRENT_RUNTIME_TOPOLOGY.md
 - MEMORY_CONTAMINATION_MAP.md
 - RUNTIME_BOUNDARY_VIOLATIONS.md
- linux-engineering/
 - README.md
- refactor/
 - CANONICAL_AUTHORITY_GRAPH.md
 - MEMORY_AUTHORITY_BOUNDARIES.md
 - MEMORY_CONTAMINATION_GRAPH.md
 - MEMORY_DEPENDENCY_GRAPH.md
 - MEMORY_SPLIT_PLAN.md
- reports/
 - FINAL_URL_HANDOFF_PATCH.md
 - PHASE_1A_GIT_VALIDATION.md
 - PHASE_2B_ROUTING_BOUNDARY.md
- governance/
 - README.md
- memory/
 - README.md
- provenance/
 - README.md
- reports/
 - linux-engineering/
 - rhcsa_existing_state_audit.md
- retrieval/
 - README.md
- runtime/
 - adaptive_routing/
 - aoia_config.json

- circadian_router.py
- config_loader.py
- deterministic_router.py
- dvm_research.md
- environment/
 - environment_router.py
 - network_patterns.md
 - traffic_profiles.json
- epistemic_kernel.py
- routing_modes.json
- stdout_logger.py
- commands/
 - __init__.py
 - base.py
 - local_commands.py
- contradiction_registry.json
- install.sh
- knowledge/
 - README.md
 - __init__.py
- bash/
 - README.md
 - skrypty-bash-podstawy.md
 - wyszukiwanie-i-filtrowanie-tekstu.md
 - zaawansowane-narzedzia-tekstowe.md
 - zmienne-rodowiskowe-i-powoka.md
- candidates/
 - candidate_command_index.json
 - candidate_commands.csv
- canonical/
 - rhcsa_commands.json
- command_graph.json
- context/
 - context_pack.json
- examples/
 - ls-command.json
 - rm-recursive-force.json
 - systemctl-status.json
- extracted/
 - linux_master_library_v1.md
 - linux_master_library_v1.txt
- filesystem/
 - README.md
 - archiwizacja-i-kompresja.md
 - edytor-vim.md
 - nawigacja-po-systemie-plikow.md
 - operacje-na-plikach-i-katalogach.md
 - przegldanie-zawartoci-plikow.md
 - wyszukiwanie-plikow.md
- index/
 - command_index.json
 - command_index_template.csv
- injection/
 - injected_context.json
- lvm/
 - README.md
 - lvm-logical-volume-manager.md

```
manifests/
  library_manifest.yaml
networking/
  README.md
  nfs-i-autofs.md
  samba-i-nfs-klient.md
  sie-konfiguracja-i-diagnostyka.md
  ssh-i-dostp-zdalny.md
  zapora-ogniowa-firewalld.md
parsed/
  rhcsa_sections.json
permissions/
  README.md
  uprawnienia-i-wasno-plikow.md
podman/
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  PROVENANCE_POLICY.md
raw/
  rhcsa_raw.txt
reports/
  category_distribution.md
  deduplication_report.md
  parsing_quality_report.md
  retrieval_engine_report.md
rhcsa_engine.py
schema/
  command.schema.json
selinux/
  README.md
  selinux.md
source/
  RHCSA_Command_Library (1).pdf
  linux_master_library_v1.pdf
storage/
  README.md
  przechowywanie-danych-dyski-i-partycje.md
  systemy-plikow-i-montowanie.md
  zarzdzanie-dyskami-raid.md
systemd/
  README.md
  boot-i-grub.md
  cron-i-harmonogramowanie-zada.md
  systemd-i-zarzdzanie-usugami.md
  zarzdzanie-pakietami-dnf-rpm.md
tools/
  CANONICAL_BUILDER_README.md
  CONTEXT_PACK_README.md
  INDEX_BUILDER_README.md
  INJECTION_LAYER_README.md
  README.md
  SECTION_PARSER_README.md
  candidate_index_loader.py
  canonical_builder.py
  context_injector.py
  context_pack_builder.py
```

- index_builder.py
- markdown_kb_builder.py
- pdf_extract.py
- section_parser.py
- troubleshooting/
 - README.md
 - diagnostyka-i-narzdzia-systemowe.md
 - dodatkowe-narzdzia-administracyjne.md
 - informacje-o-systemie.md
 - logowanie-i-monitorowanie-systemu.md
 - zarzdzanie-procesami.md
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 - zarzdzanie-grupami.md
 - zarzdzanie-uytkownikami.md
- validator/
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 - validation_report.md
 - validation_rules.py
 - validator.py
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- memory/
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 - gemma_worker_memory.py
- hats/
 - coding.json
 - linux.json
 - research.json
- rhcsa_context.py
- obsidian_vault/
 - .obsidian/
 - app.json
- 00_START_HERE.md
- Daily/
 - 2026-05-23.md
- Evidence/
 - 20260523_204053_498246.md
 - 20260523_204122_715088.md
 - 20260523_204427_843537.md
 - 20260523_204557_588315.md
- Inbox/
- Knowledge/
- Logs/
- Projects/
- Prompts/
- Reasoning/
 - 20260523_204053_498246.md
 - 20260523_204122_715088.md
 - 20260523_204427_843537.md
 - 20260523_204557_588315.md
- Sessions/
- Templates/
- orchestrator/
 - __init__.py
 - gemma_gemma.py
 - knowledge_router.py
- project_scan.json

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prompts/
  system_prompt.txt
provenance_registry.json
providers/
  __init__.py
  aureon_provider.py
  base.py
  config.py
  gemini_provider.py
  gemma_provider.py
  openai_compatible.py
requirements.txt
retrieval/
  __init__.py
  linux/
    __init__.py
    provenance_attach.py
    query_normalizer.py
    retrieval_engine.py
    scoring.py
router/
  __init__.py
  local_router.py
run.sh
run_web.sh
state/
  agent_state.json
  model_config.json
  providers.json
  token_savings_report.json
tools/
  __init__.py
  browser_tools.py
  build_rhcsa_library.py
  epistemic_registry.py
  executor.py
  filesystem_tools.py
  memory.py
  memory_hats.py
  project_scanner.py
  rhcsa_search.py
  shell_tools.py
  system_info.py
  validator.py
  web_reader.py
webapp.py
state/
  model_config.json
  providers.json
tests/
  test_aeia_determinism.py
  test_epistemic_kernel.py
  test_epistemic_registry.py
  test_epistemic_safeguards.py
  test_executor_containment.py
  test_knowledge_validator.py
  test_linux_retrieval.py
```

```
test_main.py
test_rhcsa_retrieval.py
test_routing_boundary.py
web/
```